STAT/MA 41600

In-Class Problem Set #26: October 19, 2026

1. Suppose that X and Y are continuous random variables whose joint probability density function $f_{X,Y}(x,y)$ is constant on the interior of the triangle with vertices at $(0,0)$, $(8,0)$, and $(0,5)$, and otherwise $f_{X,Y}(x,y) = 0$.

1a. Are X and Y independent? Why or why not?

1b. Find the probability density function $f_X(x)$ of X.

1c. Find the probability density function $f_Y(y)$ of Y.

2. Suppose that X and Y have joint probability density function $f_{X,Y}(x,y) = 14e^{-2x-7y}$ for $x > 0$ and $y > 0$, and otherwise $f_{X,Y}(x,y) = 0$.

2a. Are X and Y independent? Why or why not?

2b. Find the probability density function $f_X(x)$ of X.

2c. Find the probability density function $f_Y(y)$ of Y.

3. Suppose that X and Y have joint probability density function

$$f_{X,Y}(x,y) = \begin{cases} k(25-xy) & \text{if } 0 < x < 5 \text{ and } 0 < y < 5, \\ 0 & \text{otherwise,} \end{cases}$$

where k is constant.

3a. What is the value of k?

3b. Are X and Y independent? Why or why not?

3c. Find the probability density function of X.

4. Suppose that, for $0 < x < 5$ and $0 < y < 2$, we have $f_{X,Y}(x,y) = (1/25)(5-x)(2-y)$, and otherwise $f_{X,Y}(x,y) = 0$.

Are X and Y independent? Why or why not?

(We already computed the probability density function of X and the probability density function of Y on Friday.)

5. Now consider two random variables X and Y with joint probability density function $f_{X,Y}(x,y) = 63e^{-2x-7y}$ for $y > x > 0$, and otherwise $f_{X,Y}(x,y) = 0$.

5a. Are X and Y independent? Why or why not?

5b. Find the probability density function $f_Y(y)$ of Y.

(We already computed the probability density function $f_X(x)$ of X on Friday.)